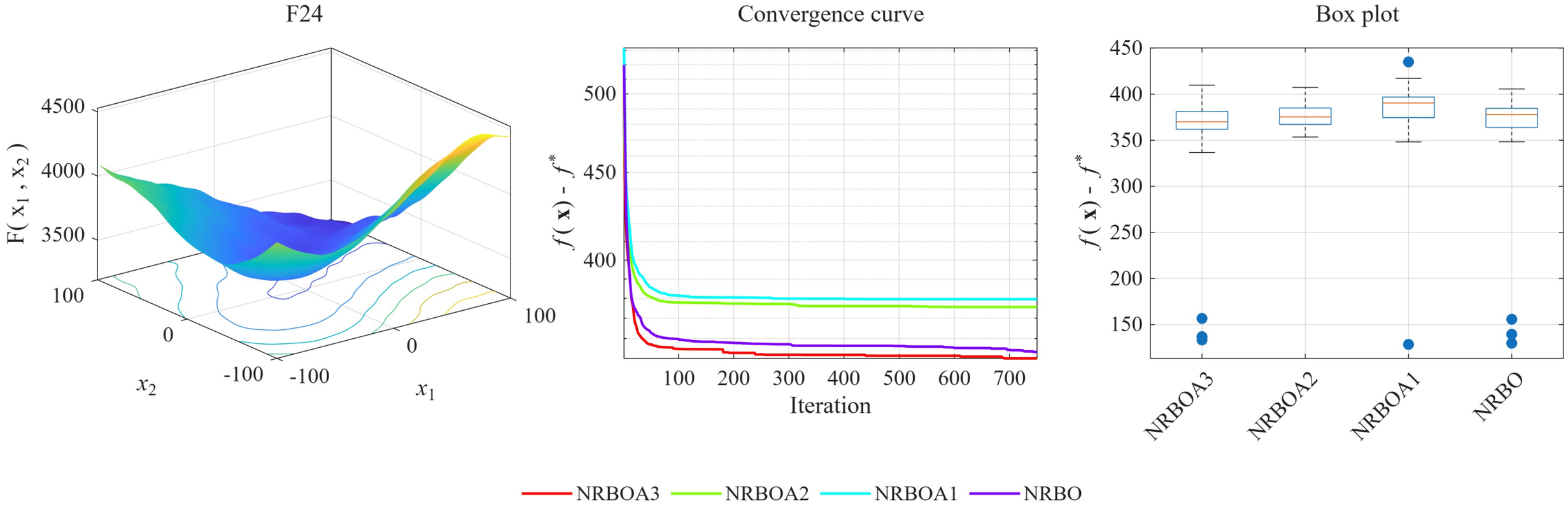

F24
4500
4000
3500
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
500
450
400
f(x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
450
400
350
300
250
200
150
f(x) - f*
NRBOA3
NRBOA2
NRBOA1
NRBO
NRBOA3
NRBOA2
NRBOA1
NRBO